

SEED STAGE — AI / E-COMMERCE INFRASTRUCTURE

ReturnShieldAI

The first AI platform built exclusively to detect and prevent e-commerce **return fraud**.

\$50B+

Annual return fraud losses

18%

YoY fraud growth

20

Modular engines

150ms

Per-request scoring

THE PROBLEM

Return fraud is the \$50B blind spot in e-commerce

Every fraud platform stops crime at checkout. Return fraud happens *after* the transaction — invisible to legacy tools.

Dimension	Payment Fraud	Return Fraud
Detection point	Checkout (pre-transaction)	Post-fulfillment (days later)
Data signals	~50 features	200+ features across order, text, graph, images
Attack surface	Card details	Physical goods, labels, weight, chat, photos
Existing coverage	Excellent (Stripe, Sift, Forter)	Near zero

THE ATTACK SURFACE

10 fraud patterns that legacy tools can't detect

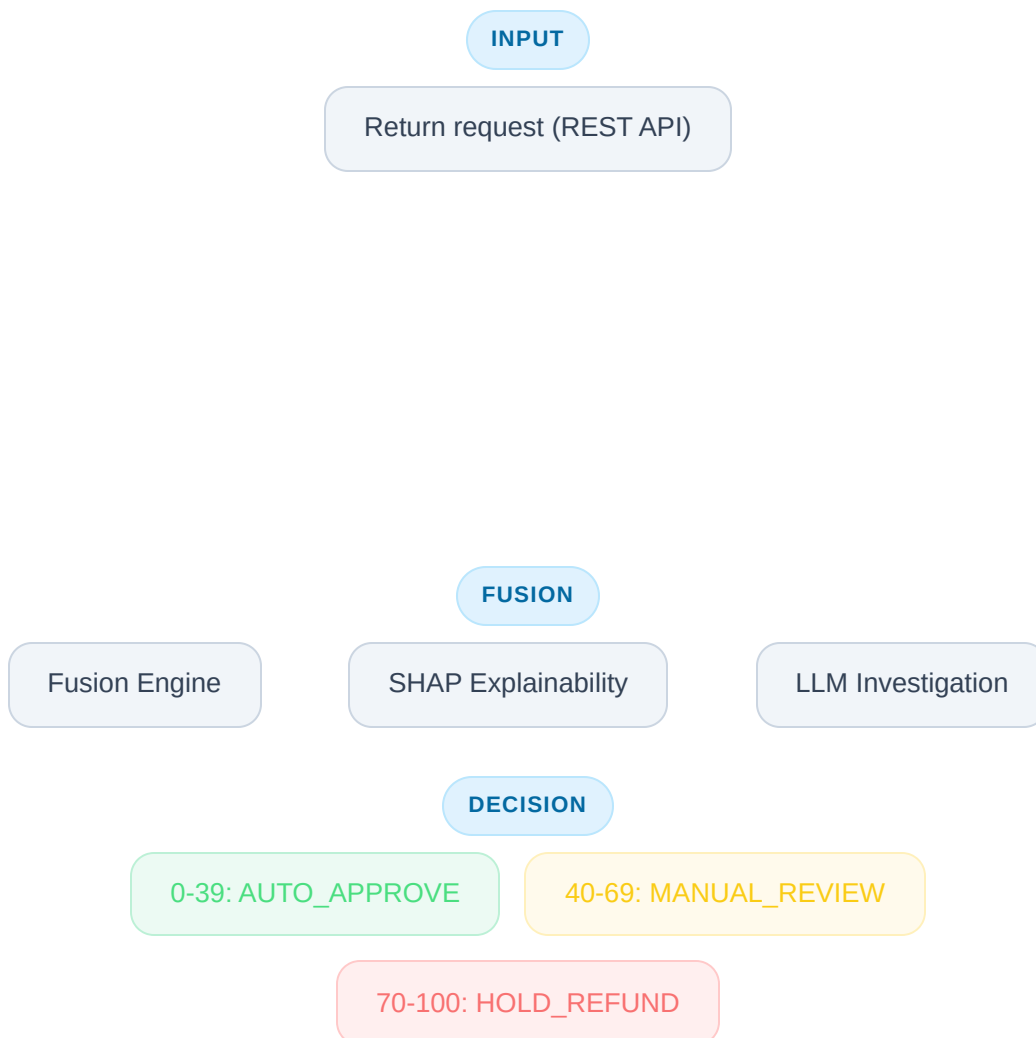
THE SOLUTION

ReturnShieldAI — **purpose-built** for return fraud

Deploy via API in 5 minutes. Every return scored, explained, and actioned before the refund is released.

ARCHITECTURE

20 engines, **one unified** decision pipeline



MARKET OPPORTUNITY

**\$48B TAM — and growing 18%
CAGR**

Target verticals

COMPETITION

Legacy vendors built for checkout

— we built for returns

Capability	ReturnShieldAI	Signifyd	Forter	Sift	NoFraud
Return fraud focus	★ Primary	★ Afterthought	★ Afterthought	★★ Partial	★★ Partial
Post-purchase detection	✓ Yes	✗ No	✗ No	◀ Limited	◀ Limited
Fraud ring detection	NetworkX + PageRank	◀ Basic	◀ Basic	◀ Basic	✗ None
NLP on return text	sentence-transformers	✗	✗	◀	✗
Explainability	Full SHAP + LLM	◀◀	◀◀	◀◀	◀
Self-hosted option	✓ Yes	✗ Cloud-only	✗ Cloud-only	✗ Cloud-only	✗ Cloud-only
Time to deploy	Minutes	Weeks	Weeks	Days	Days

TRACTION

Production-ready prototype — built in weeks

- ✓ **20 modular engines**
Built, tested, and integrated end-to-end
- ✓ **24 REST API endpoints**
Full CRUD, scoring, dashboard, module management
- ✓ **Production frontend**
React + TypeScript + Tailwind, built with Vite
- ✓ **3 Kaggle datasets ingested**
2M+ rows across e-commerce domains
- ✓ **5 ML model families**
Structured (RF/LGBM/XGB), NLP, anomaly, fusion, graph
- ✓ **Feedback loop**
Analyst decisions retrain models continuously

BUSINESS MODEL

Usage-based pricing for **every** **scale**

Per-return pricing

\$0.05–0.10

Implementation fee

\$5K–25K

Revenue model

**Usage + % of
fraud
prevented**

COMPETITIVE MOAT

Why we win — 5 structural advantages

TEAM

**Built by engineers who understand
fraud at scale**

THE ASK

Seed round: \$750K

Engineering (3 FTE, 12 months)	\$412K	55%
Customer pilots (3-5 merchants)	\$112K	15%
Cloud infrastructure	\$90K	12%
Compliance & legal (SOC 2)	\$75K	10%
Marketing & GTM	\$60K	8%
Total	\$750K	100%

TARGET INVESTORS

Enterprise SaaS / fintech infrastructure VCs · E-commerce & logistics funds · AI/ML specialists · Strategic angels from Stripe, Shopify, Forter

LET'S BUILD THE FUTURE OF FRAUD PREVENTION

ReturnShieldAI

The \$50B return fraud problem isn't going away.

Help us build the first platform that solves it.

Contact

senthilnath@returnshield.ai

GitHub

github.com/senthilnathang/ReturnShieldAI